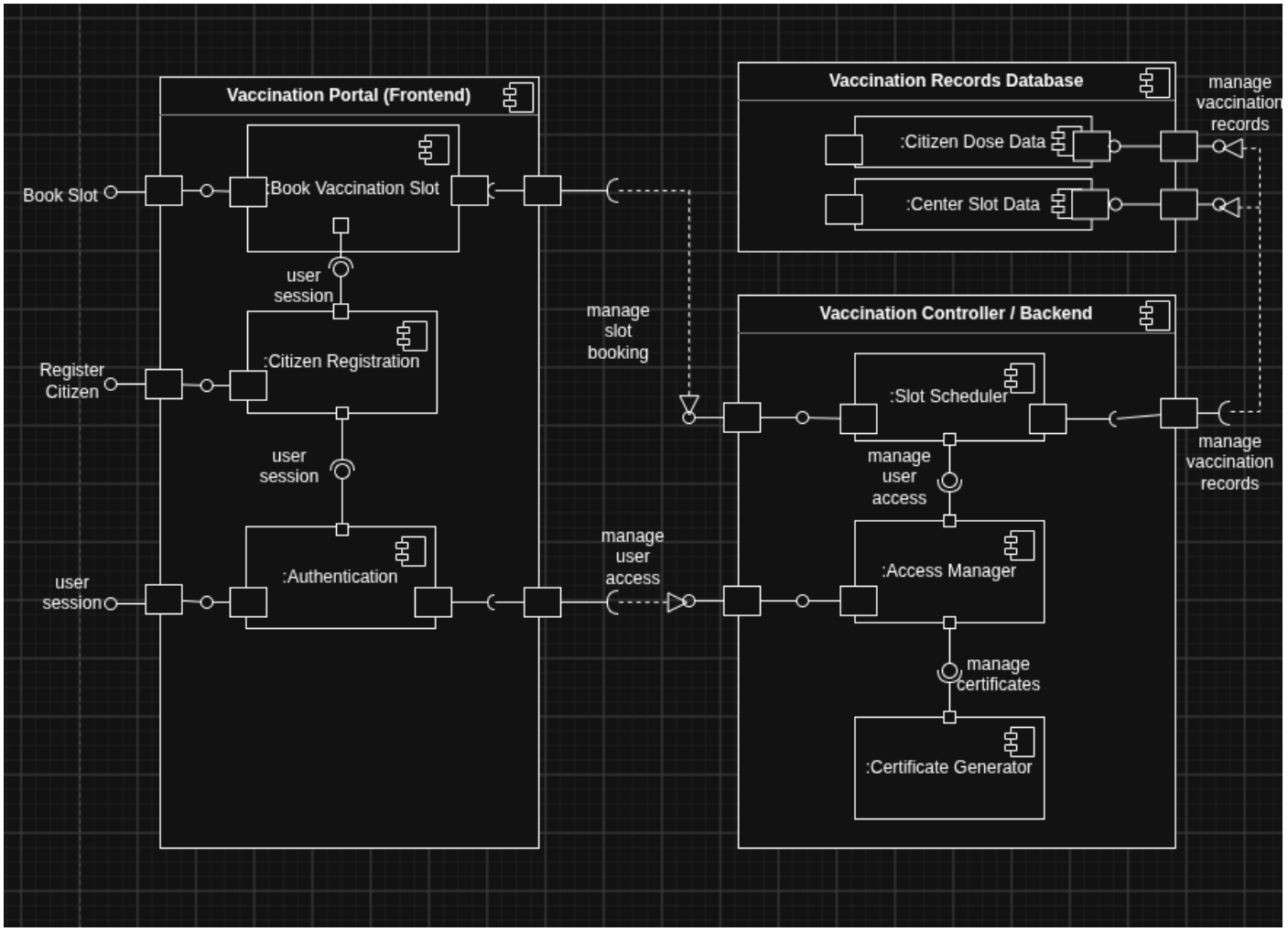
SHALIK S | PES2UG24CS454

Lab 3



1. Components

Component	What it does
Vaccination Portal (Frontend)	Subsystem holding the citizen and officer facing screens of the vaccination system.
:Book Vaccination Slot	Screen through which a citizen picks a centre, date and slot and submits the booking request.
:Citizen Registration	Captures the citizen's details and registers them into the correct vaccination cohort.
:Authentication	Signs citizens and officers in and holds the session used by the other frontend modules.
Vaccination Controller / Backend	Subsystem holding the business logic for eligibility, scheduling and certificates.
:Slot Scheduler	Checks the dose interval and slot availability, then confirms the booking and updates the schedule.
:Access Manager	Validates the session and decides which citizen or officer actions are allowed.
:Certificate Generator	Records the administered dose and issues the digitally signed QR vaccination certificate.
Vaccination Records Database	Subsystem that persists all vaccination data used by the backend.
:Citizen Dose Data	Stores citizen profiles, cohort assignment and the dose history of each citizen.
:Center Slot Data	Stores vaccination centres, slot capacity and the bookings made against each slot.

2. Interfaces & Relationships

Interface / Relationship	Provided by → Required by
Book Slot	Provided by :Book Vaccination Slot to the citizen; public entry point for booking a dose.
Register Citizen	Provided by :Citizen Registration to the citizen; public entry point for cohort registration.
user session	Provided by :Authentication, required by :Book Vaccination Slot and :Citizen Registration for a logged-in session.

manage slot booking	Provided by :Slot Scheduler, required by :Book Vaccination Slot for eligibility check and slot confirmation.
manage user access	Provided by :Access Manager, required by :Authentication to validate sessions and roles.
manage certificates	Provided by :Certificate Generator, required by the portal to record a dose and fetch the QR certificate.
manage vaccination records	Provided by :Citizen Dose Data and :Center Slot Data, required by the backend modules for all reads and writes.